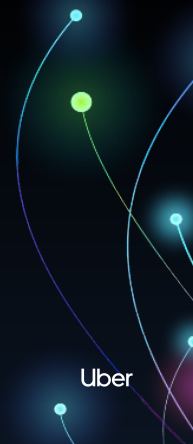




Modernizing Artifact Storage at Uber

Preetam Dwivedi
Manjari Akella



About Us



Preetam Dwivedi

Staff Engineer,
Developer Platform,
Uber



Manjari Akella

Sr. Engineer,
Developer Platform,
Uber

Outline

- Engineering at Uber
- Artifactory Concepts
- Onprem Architecture
- Cloud Architecture
- Results, Trade-offs & Learnings
- Q&A

Engineering **at Uber**



On Our Way

Uber Engineering at Scale

Development Scale

5K+

Engineers

36K+

Commits / Week

3M+

Builds / Week

440M+

Lines of Code

Monorepo Stats

Go

35 GB

- 227M+ LOC
- 22K commits/wk

Java

17 GB

- 57M+ LOC
- 3.4K commits/wk

iOS

9 GB

- 20M LOC
- 1.2K commits/wk

Web

18 GB

- 81M LOC
- 4K commits/wk

Python

11 GB

- 40M LOC
- 3.2K commits/wk

Android

7 GB

- 16M LOC
- 1.8K commits/wk

Artifactory at Uber Scale

Scale & Volume Metrics



200M

Requests Per Day



7TB

Uploaded / Month



7PB

Downloaded / Month



145TB

Total Storage

Supported Packages and Registries



Docker



npm



Maven



PyPI



Hugging Face



Gradle



Go



Debian



RPM



Swift



Terraform /
OpenTofu



Agent
Packages



Alpine



Ansible



Bower



Cargo



Chef



CocoaPods



Conan



Conda



CRAN



OCI



Machine
Learning



Nix



RubyGems



GitLfs



Helm



Ivy



LuaRocks



Skills



NuGet



Opkg



Composer
php



Pub / Dart



Puppet



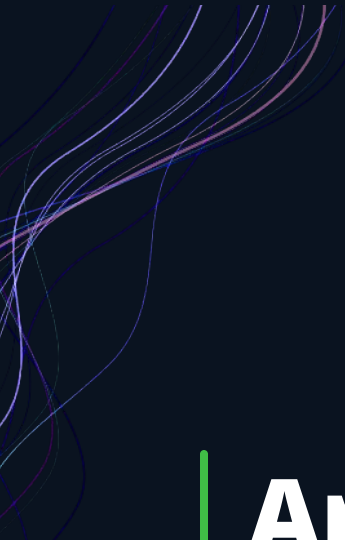
SBT



Vagrant

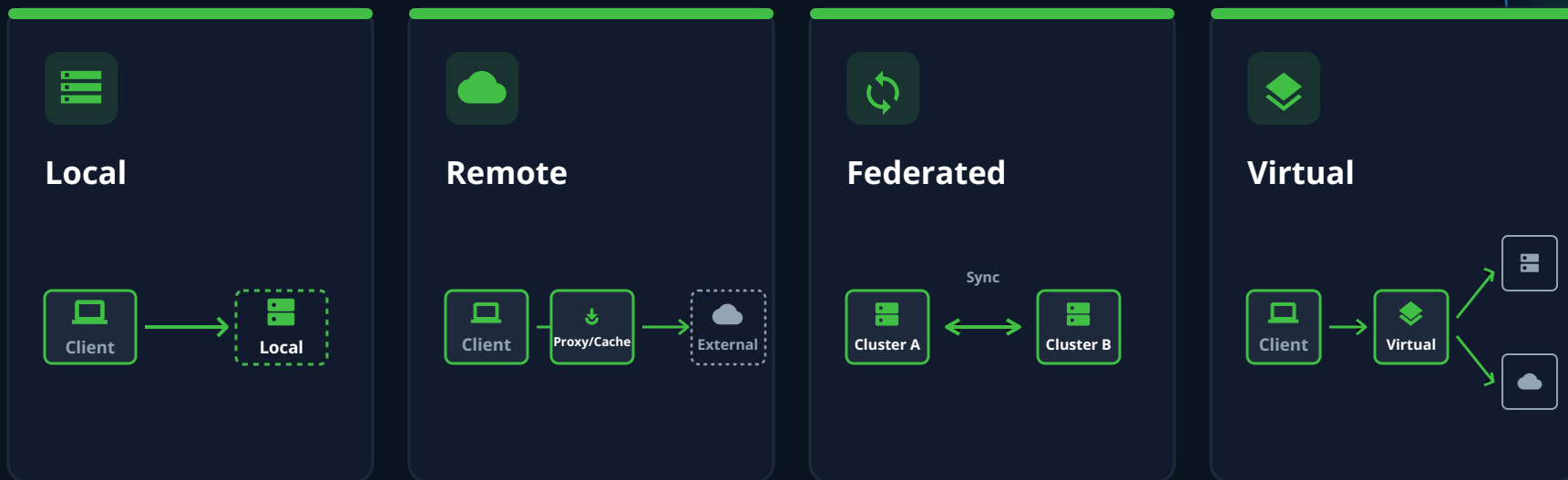


Generic

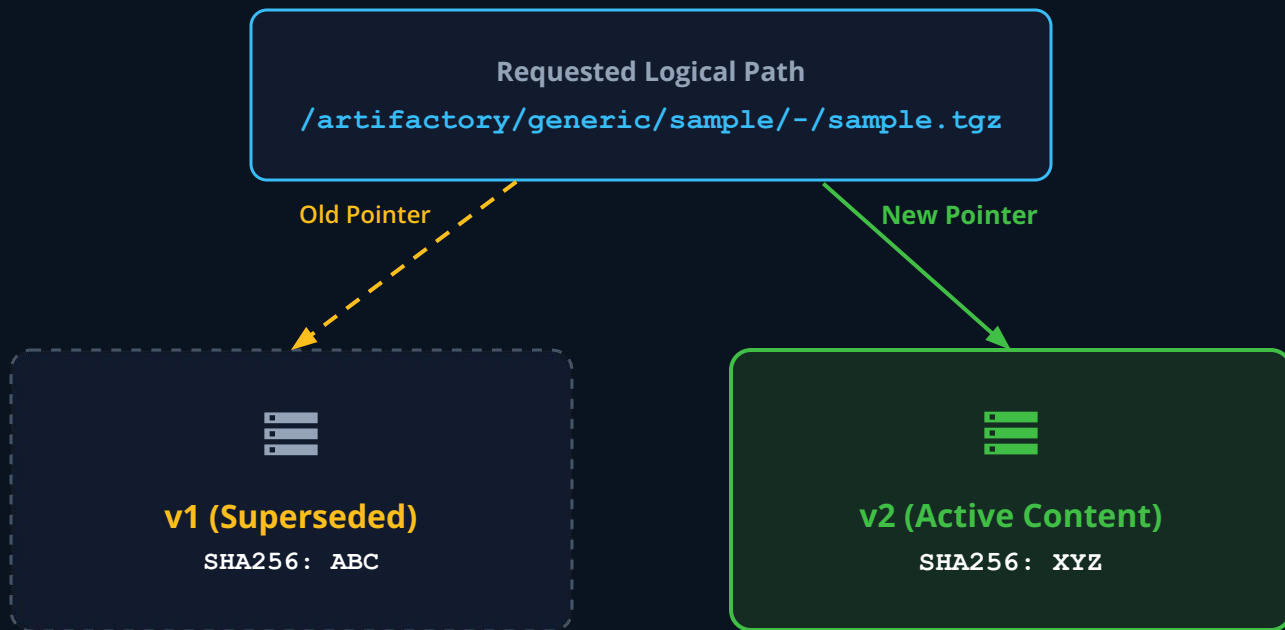


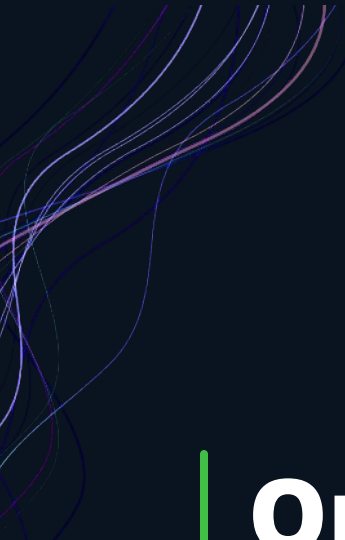
| **Artifactory Concepts**

Repository Types



Artifacts are content addressable





| Onprem Architecture

Onprem Architecture



Active-Active Cluster

High-availability deployment across active data centers.



Artifacts on Local Disk

Stored directly on local SSDs with node-level redundancy.



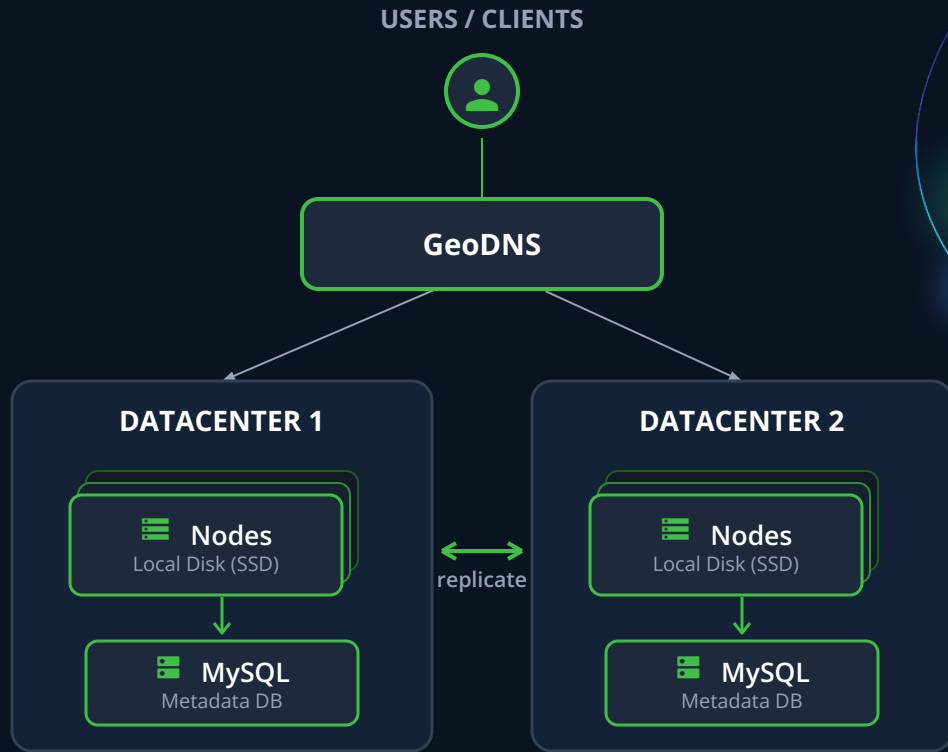
Optimistic Quorum

Accepts uploads and replicates asynchronously to meet quorum without blocking clients.

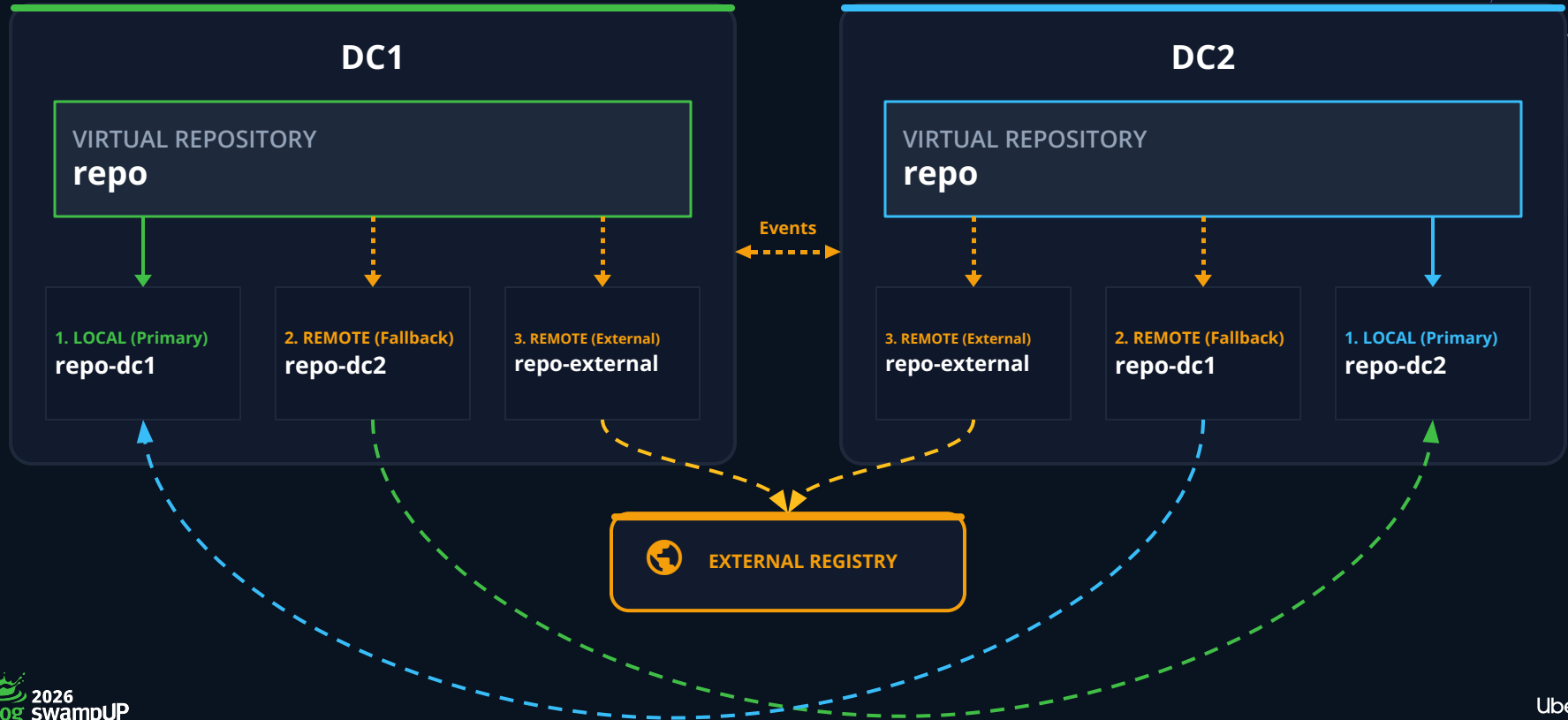


Pull Replication

Event and cron based pull replication



Onprem Repository Structure



Challenges of the Onprem Architecture



**Disk Space
Constraints**



**Hardware
Failures**



Risk of Data Loss

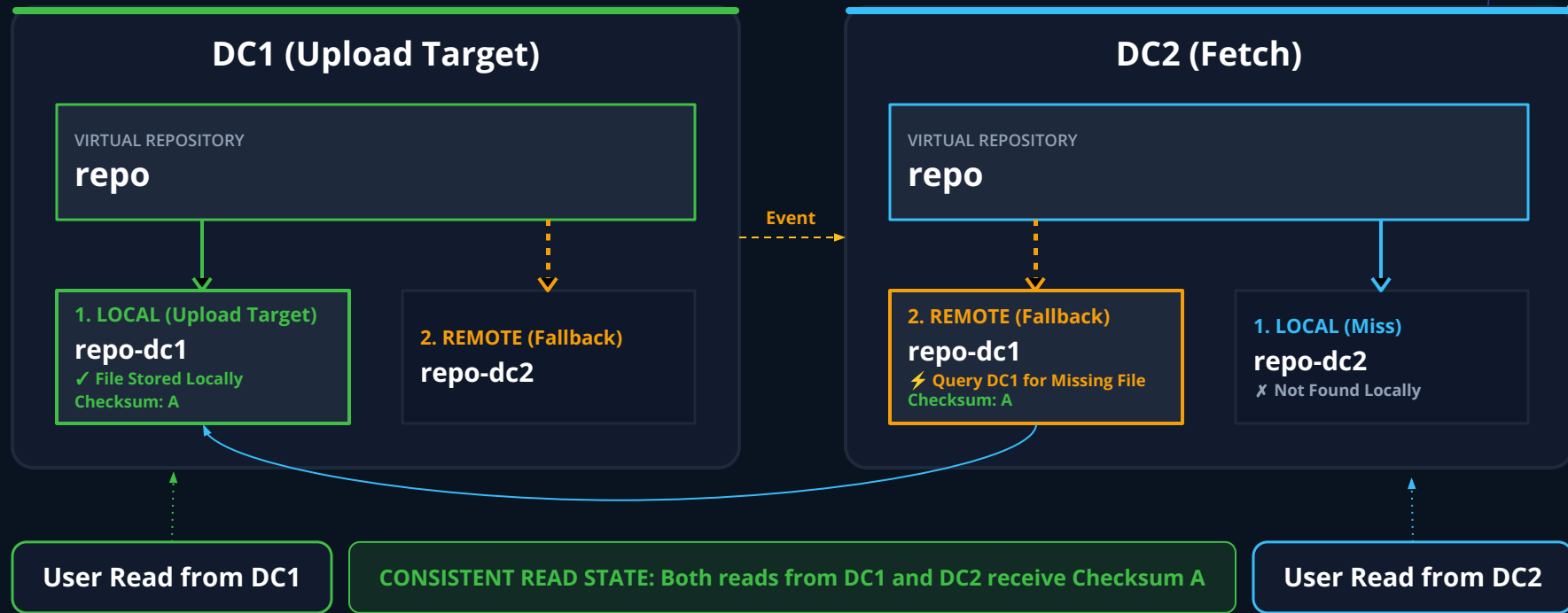


**Upgrades /
Patches**

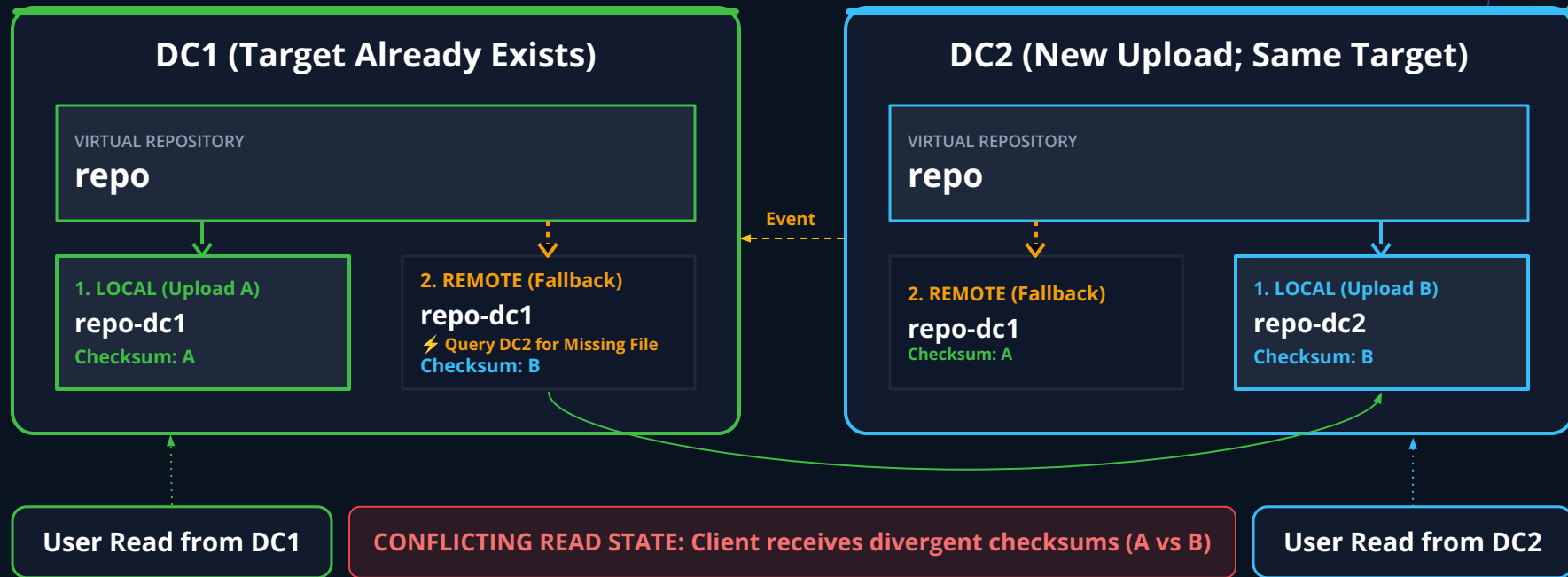


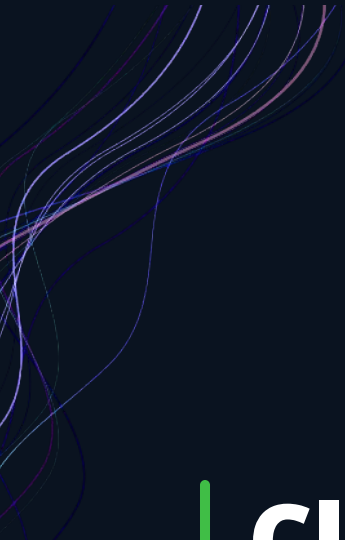
Divergent State

Example: Target Uploaded to DC1



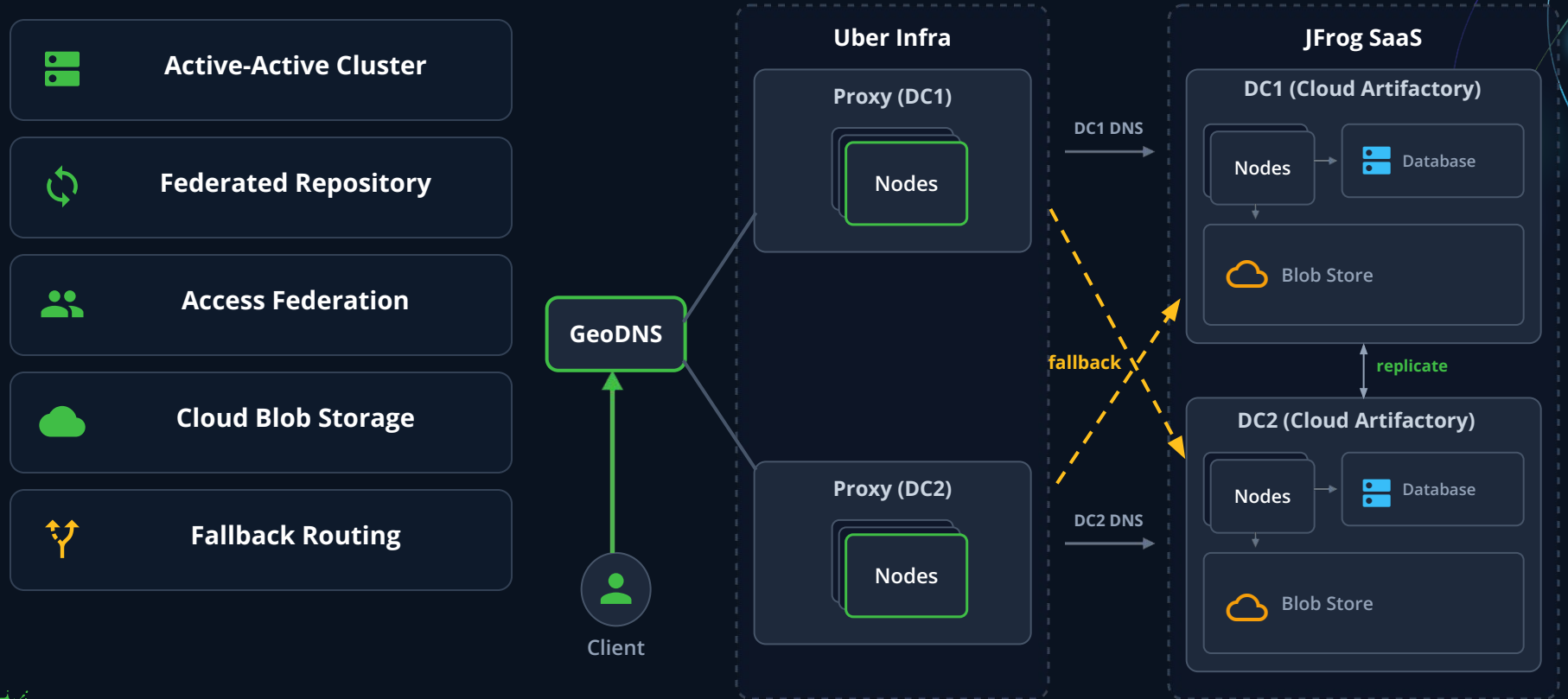
Example: Target Upload with Different Content from DC2



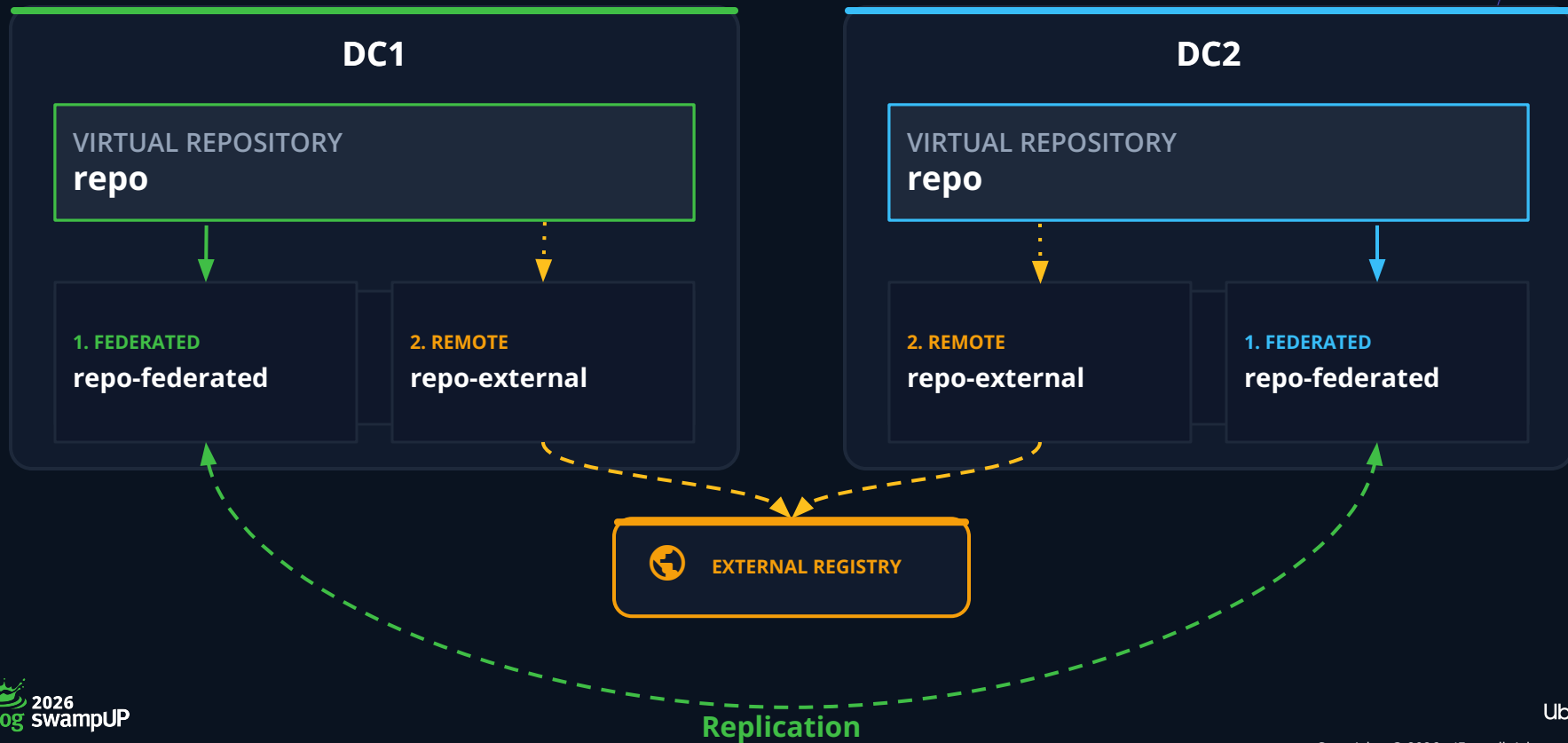


| Cloud Architecture

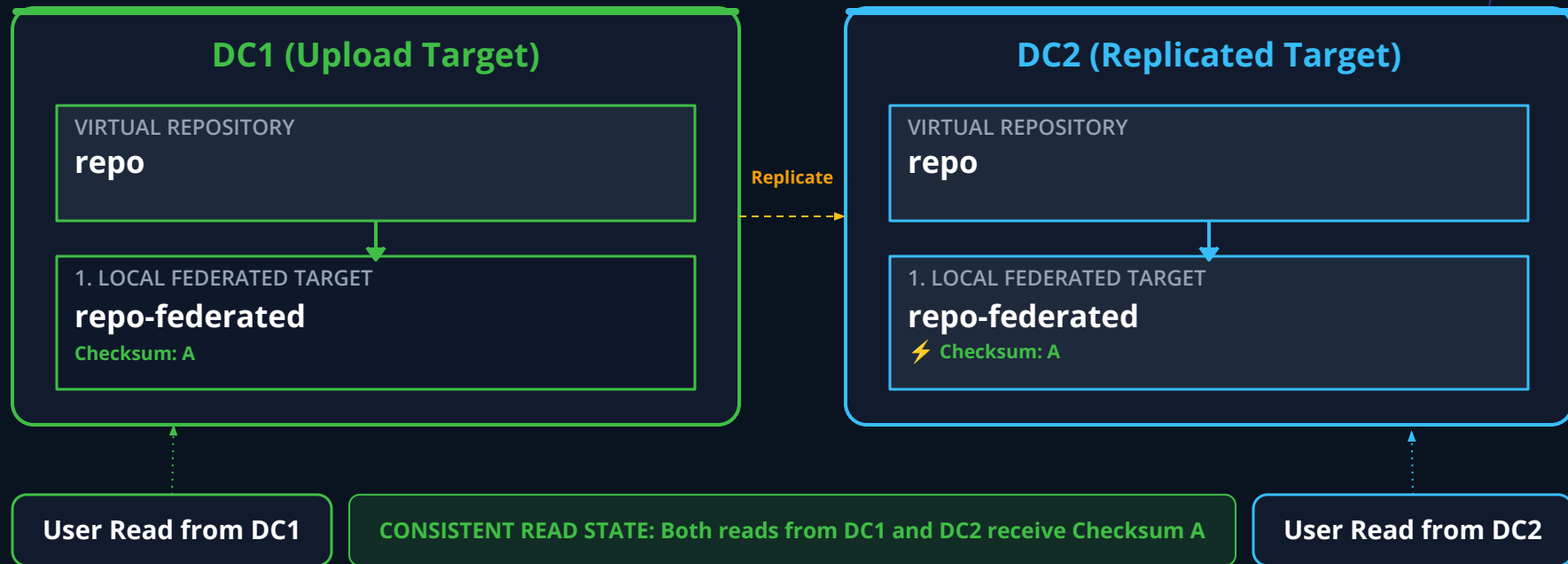
JFrog Cloud (SaaS) Architecture



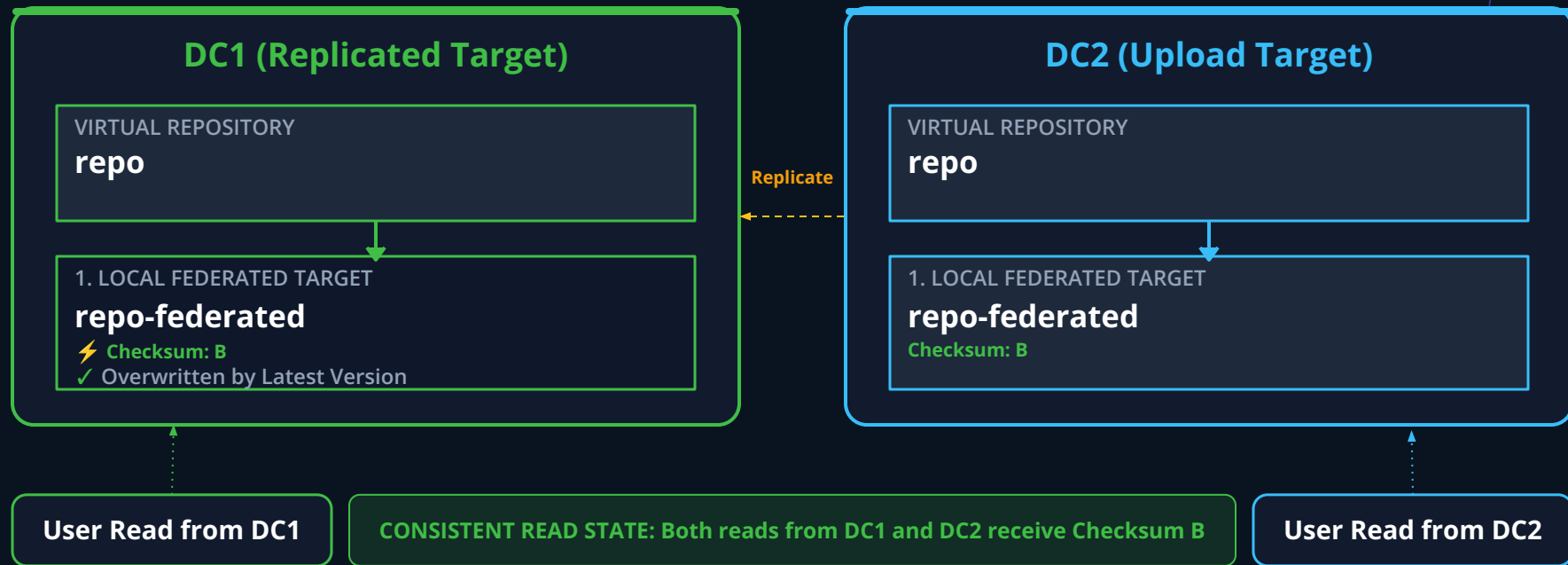
Cloud Repository Structure

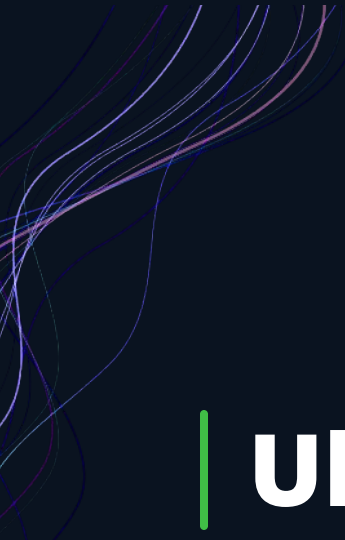


Example: Target Uploaded to DC1



Example: Target Upload with Different Content from DC2





| Uber's Proxy

Core Proxy Features



Authentication & Identity

Manages short-lived session tokens and in-memory storage.



Tracing & Observability

Provides deep request visibility, logging, and metrics.



On-Prem to Cloud

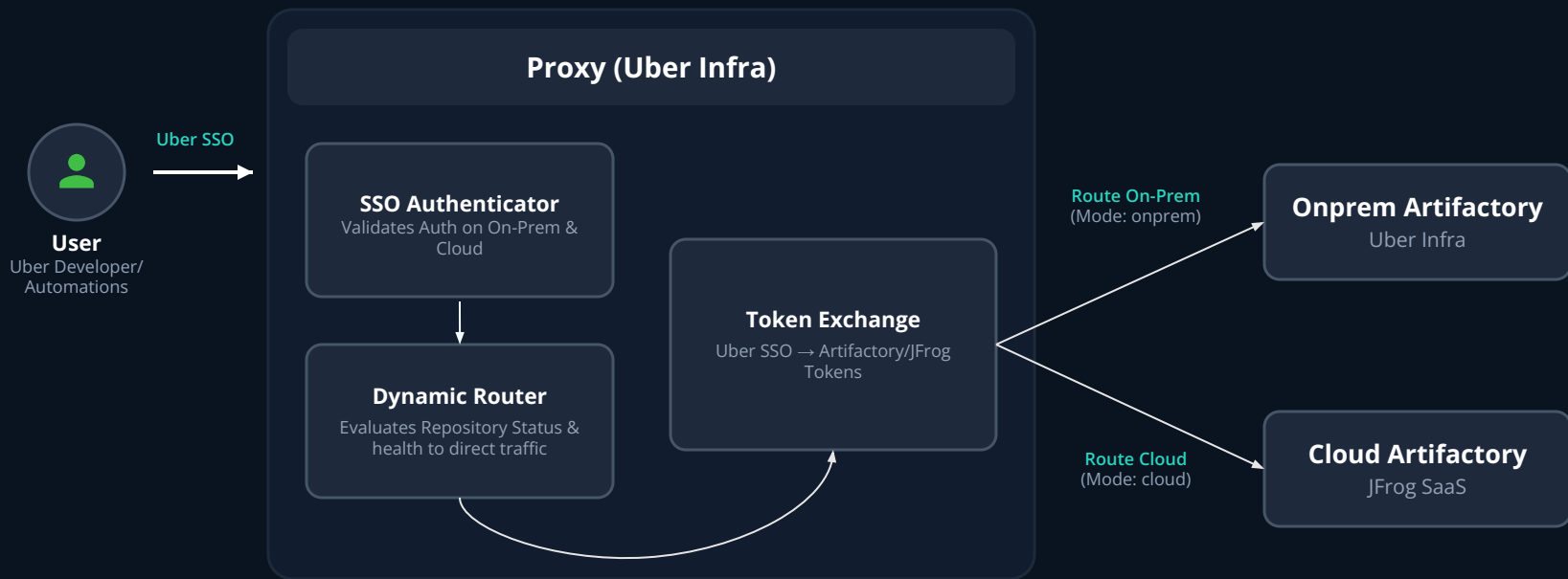
Enables seamless and secure transitioning from on-prem environments.



Failover Routing

Enables dynamic, resilient traffic redirection during outages.

User Authentication and Identity Forwarding



Core Proxy Features



Authentication & Identity

Manages short-lived session tokens and in-memory storage.



Tracing & Observability

Provides deep request visibility, logging, and metrics.



On-Prem to Cloud

Enables seamless and secure transitioning from on-prem environments.



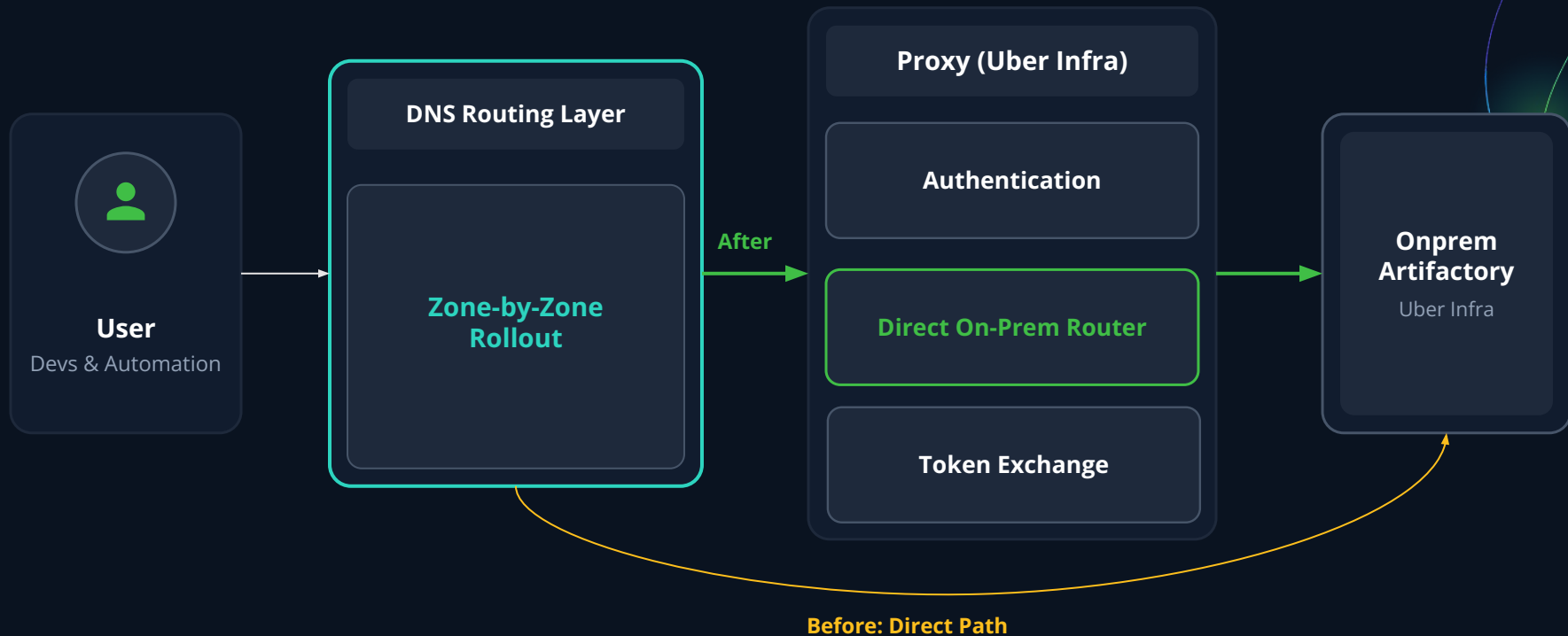
Failover Routing

Enables dynamic, resilient traffic redirection during outages.

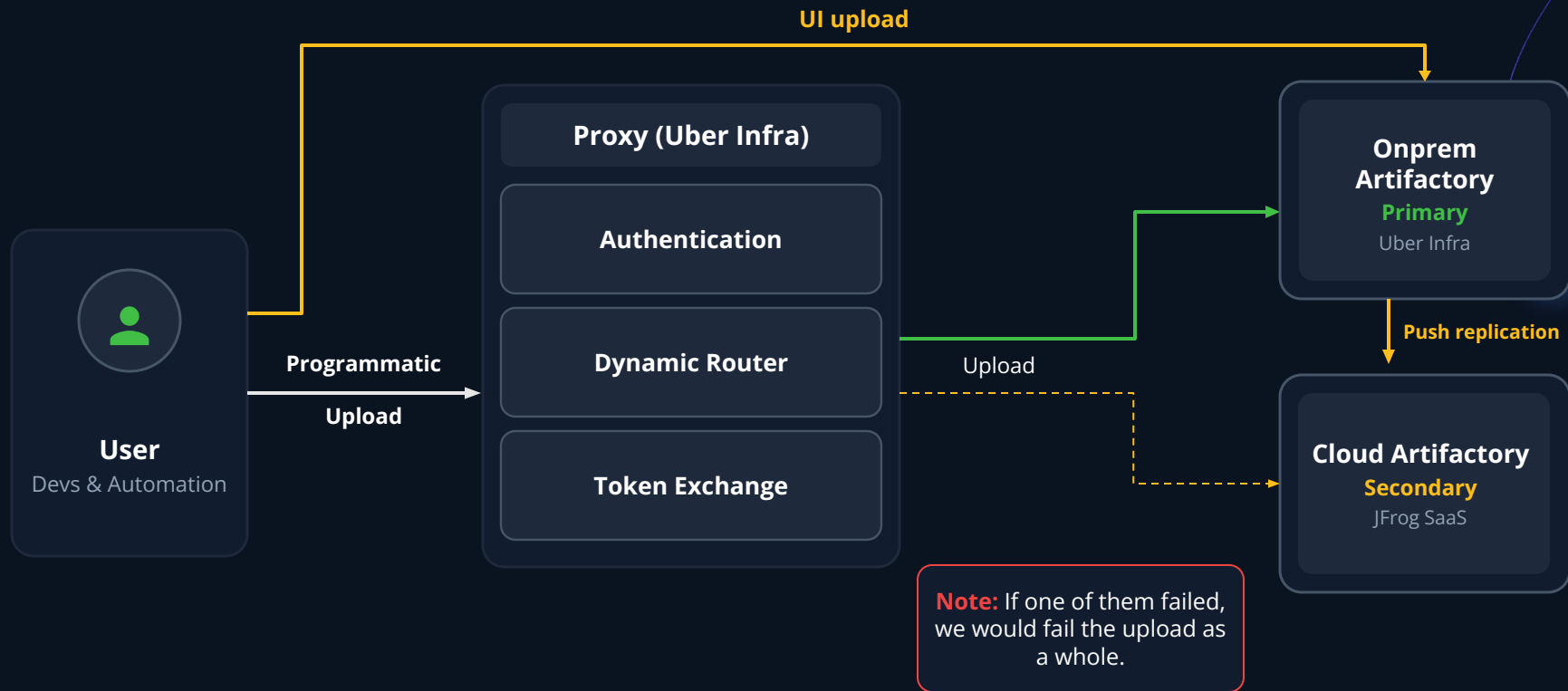
Data Migration



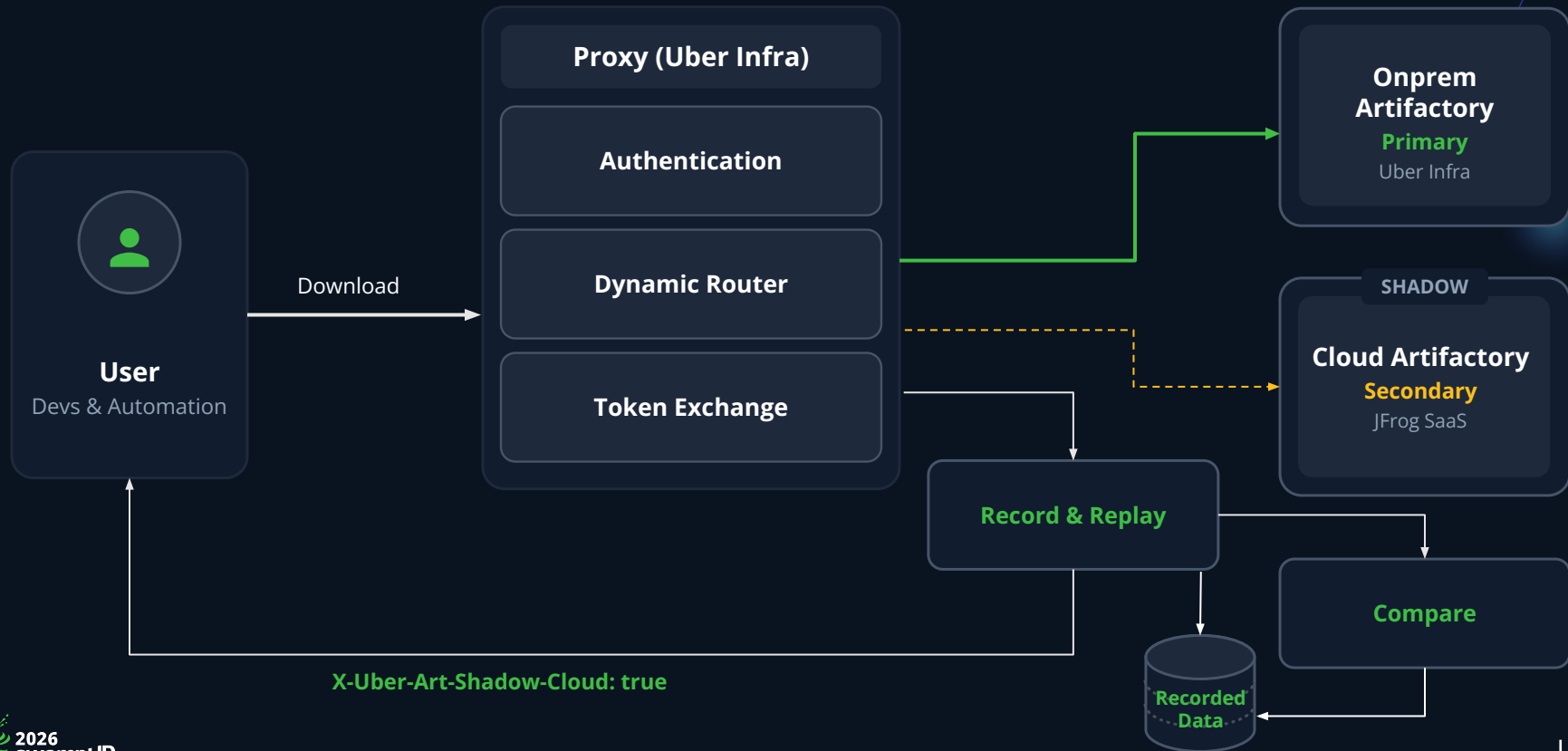
Onprem behind Proxy Rollout



Dual Write Rollout



Cloud Shadow Rollout



Correctness Verification for Shadow testing



Status Code

Validates HTTP response status codes match exactly between environments.



Checksum

Verifies and matches cryptographic checksums of transferred artifacts.



Artifact Size

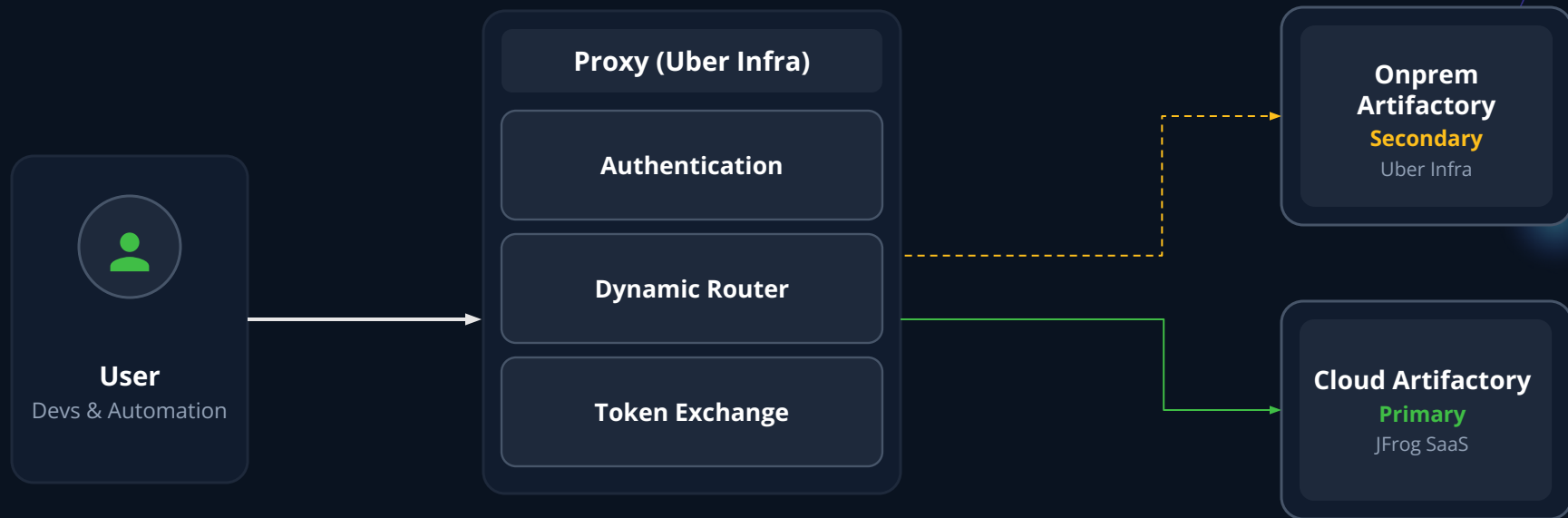
Compares declared HTTP Content-Length headers and ensures raw response byte sizes are precisely identical.



Response Bytes

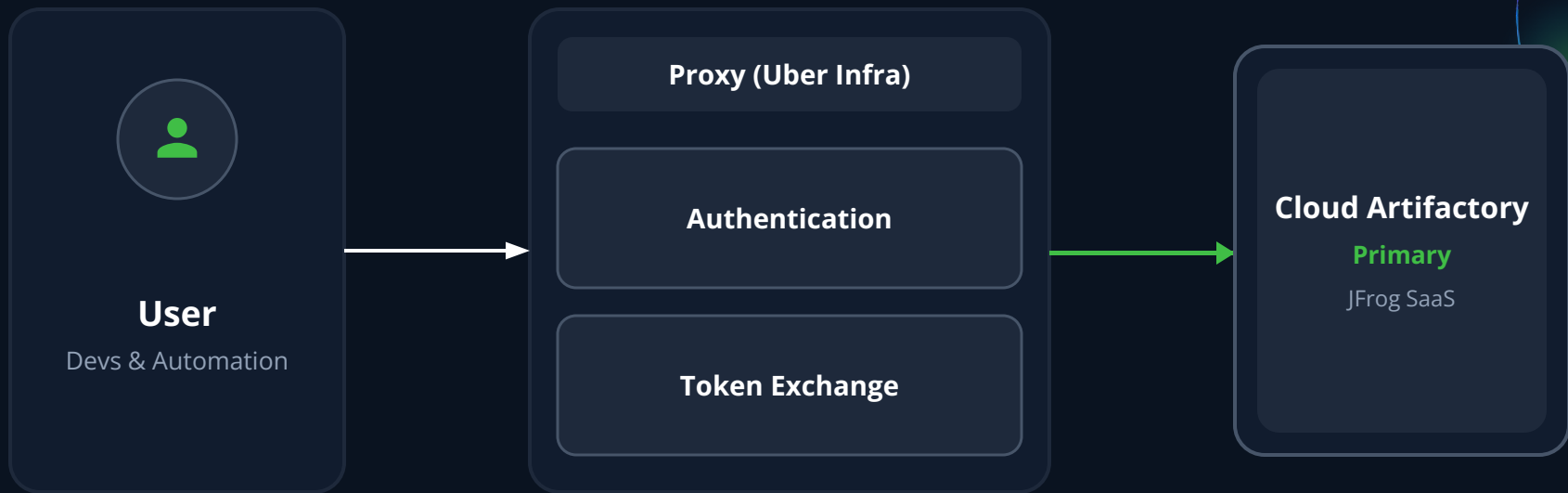
Deep byte-by-byte comparison executed for text-based artifact use cases.

Cloud Rollout



Rollout Strategy: Gradual rollout executed on a per-repository basis, prioritizing low-traffic repositories first, and then systematically migrating all remaining repositories.

End State



Core Proxy Features



Authentication & Identity

Manages short-lived session tokens and in-memory storage.



Tracing & Observability

Provides deep request visibility, logging, and metrics.



On-Prem to Cloud

Enables seamless and secure transitioning from on-prem environments.



Failover Routing

Enables dynamic, resilient traffic redirection during outages.

Observability



Error Classification

Categorizes system errors in real-time across internal services and external APIs.



Request Latency

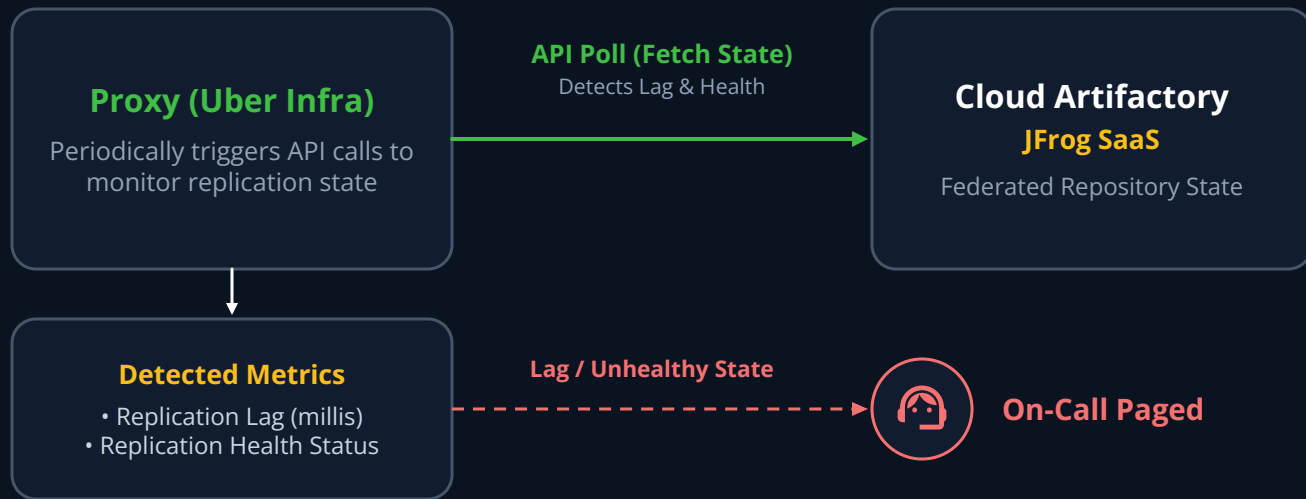
Provides granular, hop-by-hop latency breakdowns at the 95th and 99th percentiles.



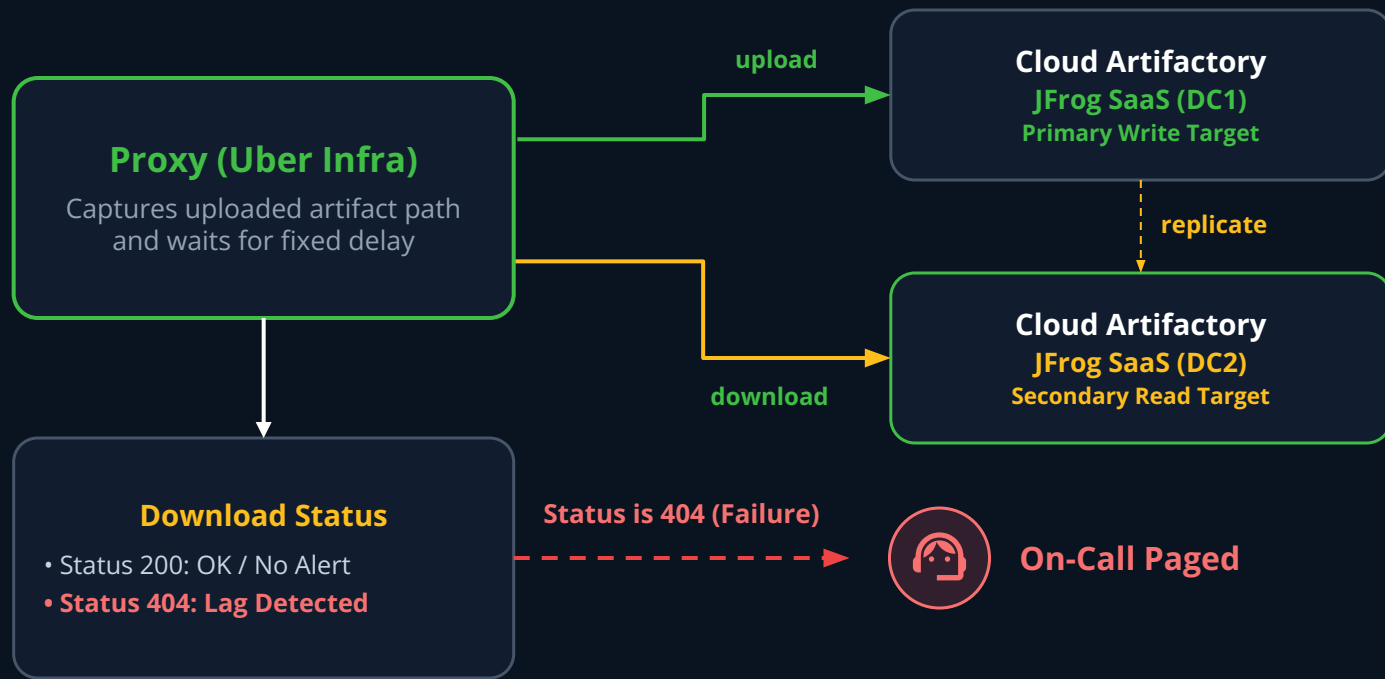
Lag Detection

Monitors replication delay across environments and automatically triggers on-call pages.

Federated Repository Lag Detection: JFrog API



Federated Repository Lag Detection: Read after write



Core Proxy Features



Authentication & Identity

Manages short-lived session tokens and in-memory storage.



Tracing & Observability

Provides deep request visibility, logging, and metrics.



On-Prem to Cloud

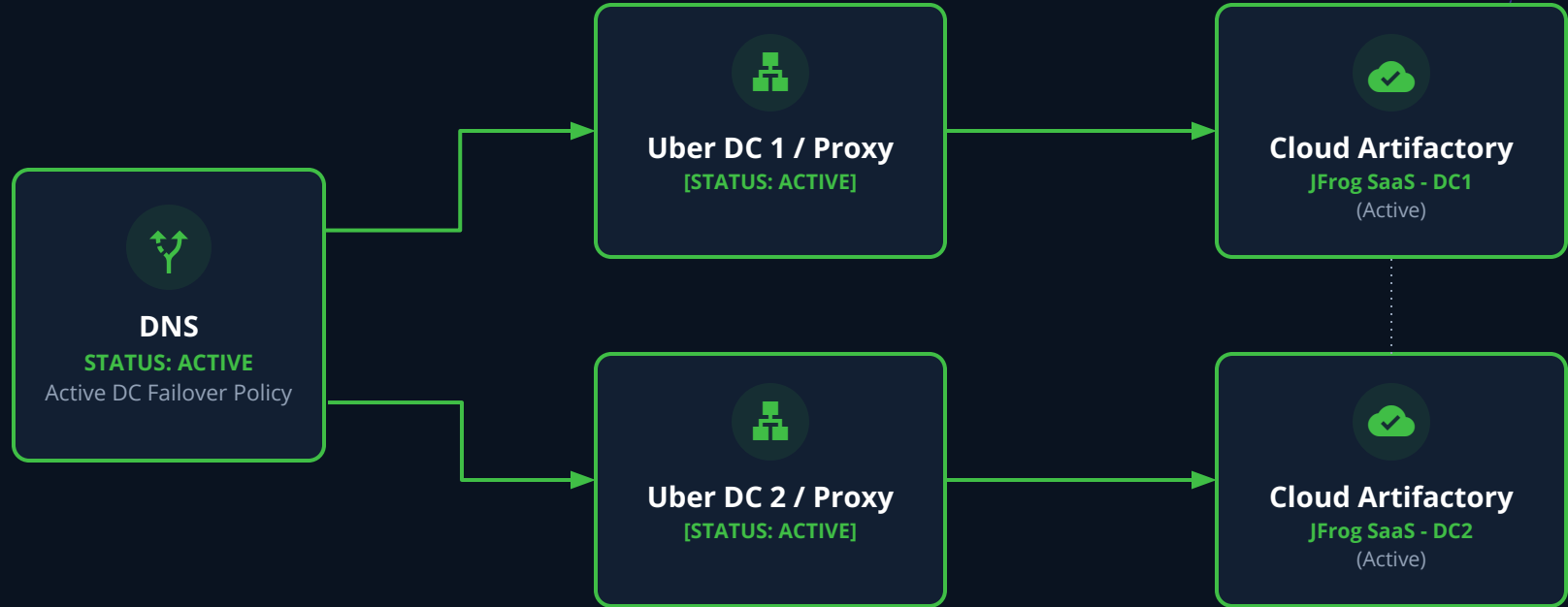
Enables seamless and secure transitioning from on-prem environments.



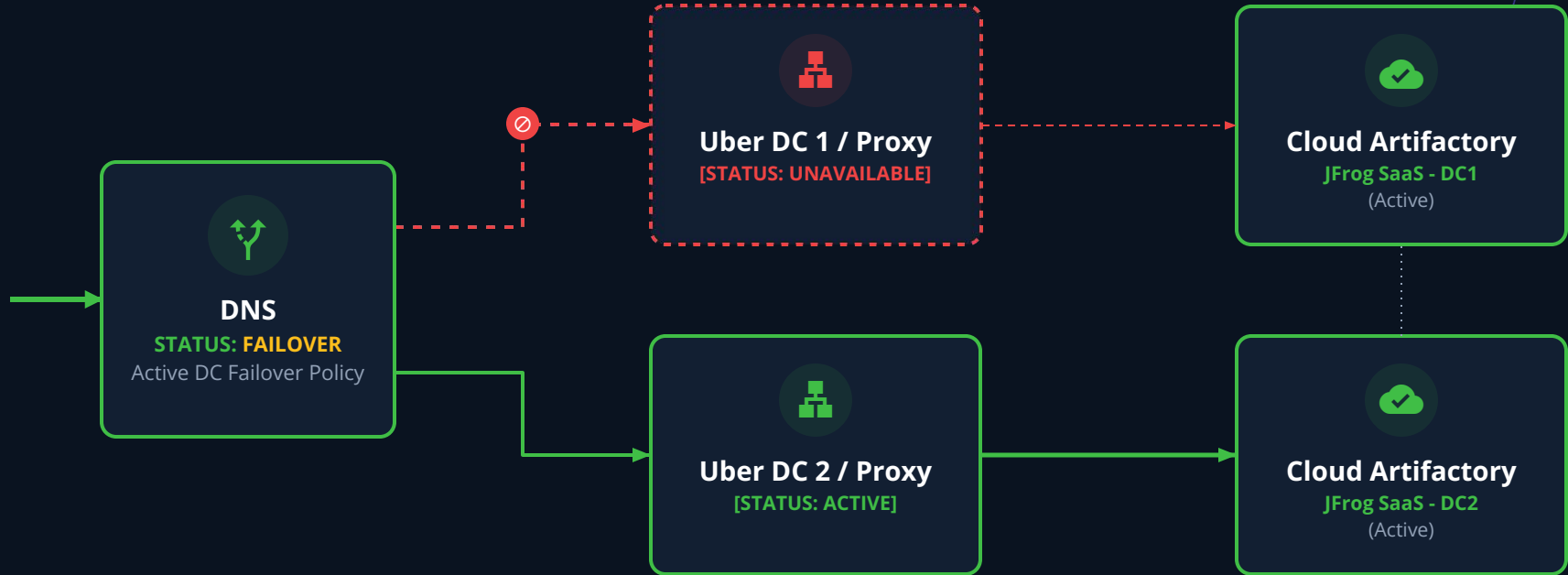
Failover Routing

Enables dynamic, resilient traffic redirection during outages.

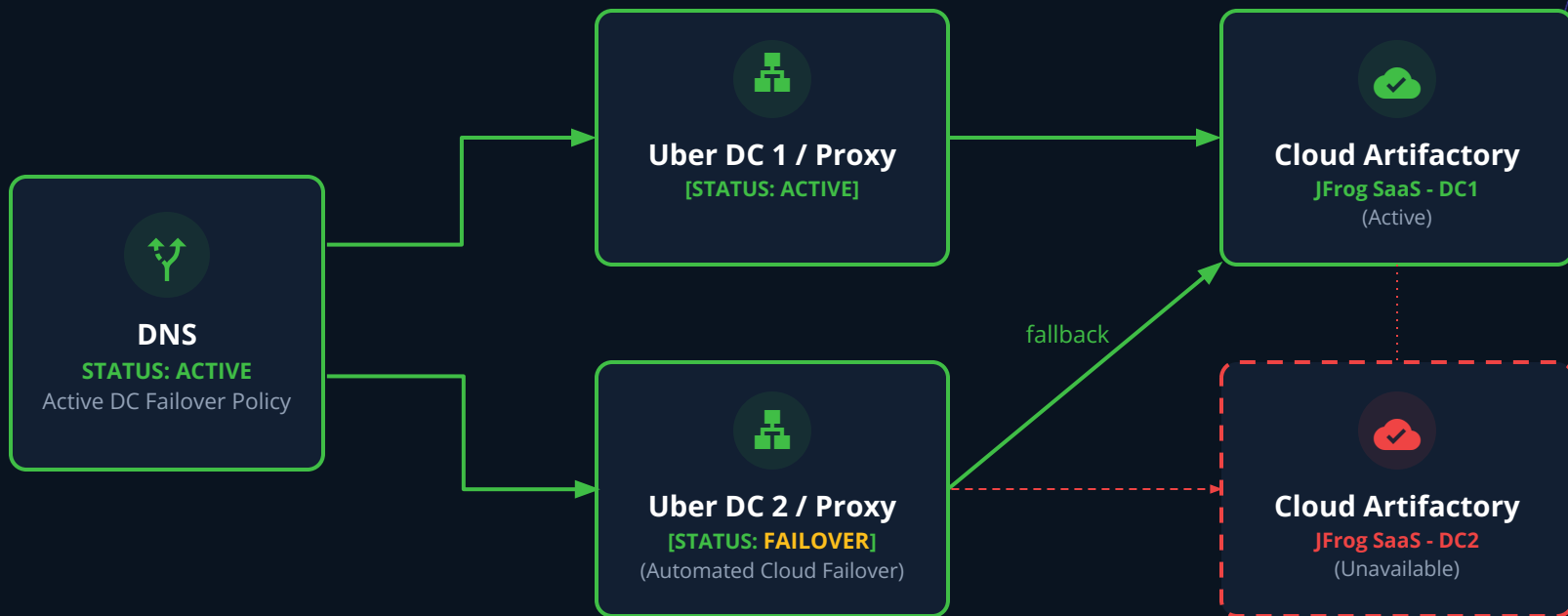
Disaster Recovery: Happy Path



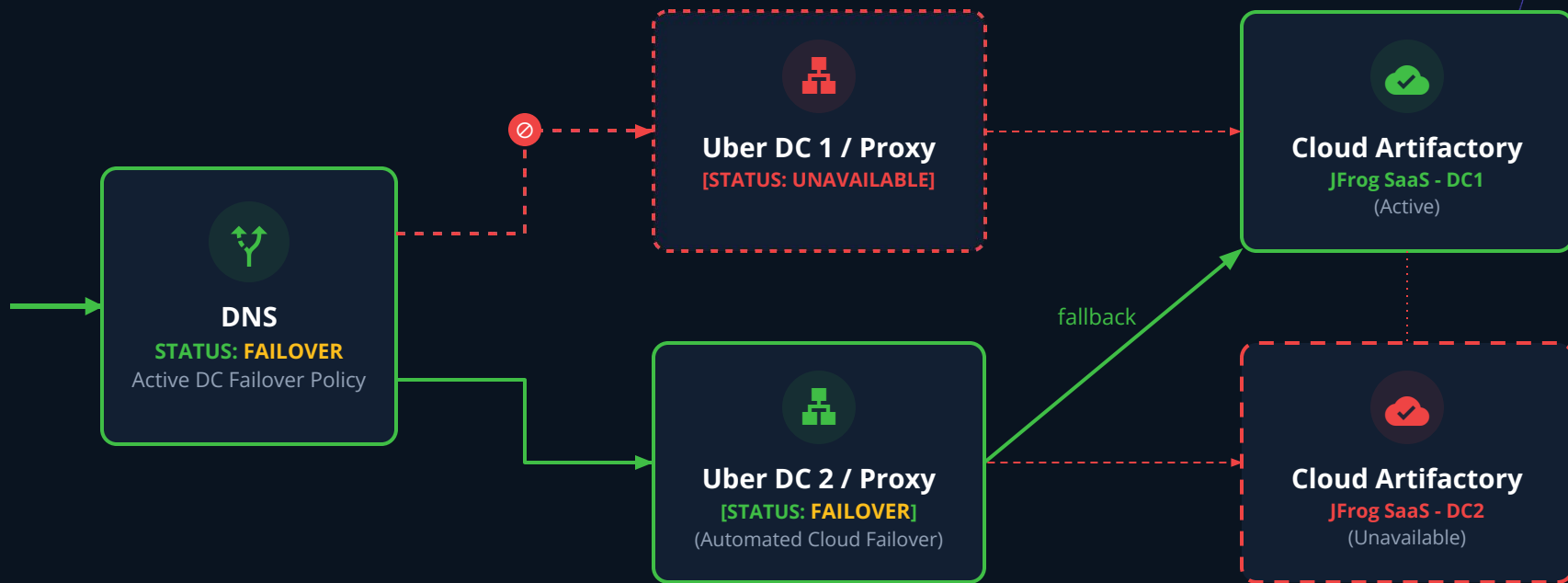
Disaster Recovery: Uber Data Center Failover

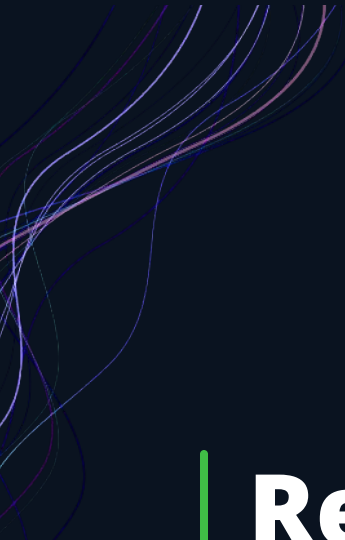


Disaster Recovery: JFrog Cloud DC Failover



Disaster Recovery: Uber and JFrog Cloud DC Failover





Results, Trade-offs and Learnings

Results & Impact



Availability

ON-PREM:
Not Measured

CLOUD:
99.99%



Latency

ON-PREM:
Low Latency

CLOUD:
Comparable



Visibility

ON-PREM:
Minimal

CLOUD:
Full Coverage



Dev Exp

ON-PREM:
High Friction

CLOUD:
Improved

Trade Offs



Uber Infrastructure

Natural Integration

Leverages existing monorepo & deployment infrastructure seamlessly.

Zero-Touch Upgrades

Platform upgrades happen naturally with no manual steps required for the team.



Integration & Overhead

Auth Complexity

Requires additional integration work and special Auth handling for Docker and Terraform.

Operational Learning

Team must learn proxy capabilities including authentication, and registry mechanics.



JFrog SaaS Benefits

- ✓ Access Federation
- ✓ Curation
- ✓ Federated Repos
- ✓ Upgrades & Patches

Key Learnings



Shadow Validation

Universal Concept

Implementing shadowing has proven invaluable for validating production traffic structures safely.



Future Paradigm

Carry Forward

Established as a core engineering paradigm carried forward to multiple newer software projects today.



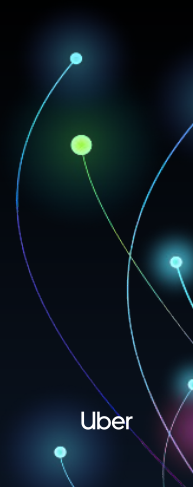
Registry & Auth

Secure External Ingress

Overcoming orchestration bottlenecks and handling token translation for complex external Docker pulls required implementing a robust multi-request authentication setup.



Thank You



Q&A

